Week 1:

1. What kind of software & hardware are/will be on our drone? (Summary of the next 5 weeks)
    a. The drone will have flight controllers, WiFi antennas, a GPS receiver, an onboard computer, and a camera. Drones will also have object detection and aerial image stitching capabilities.
2. What kinds of challenges/projects can YOU work on as a Software team member?
    a. I would like to work on AI vision and autonomous projects as a member of the software team.
3. What's the current state of drone autonomy software?
    a. The drone industry is moving towards using multiple drones with one operator working together and beyond visual line of sight, which has a drone outside the direct, unaided visual range of the remote pilot or visual observers.
4. How does one learn about drones effectively in 2026?
    a. One can learn about drones effectively by discussing with AI and reaching out to others in both the software subteam and those outside the software subteam.

Week 2:

1. How are drones used for aerial imagery purposes?
    a. Drones are used for agriculture, law enforcement, environmental purposes, and search and rescue.
2. How is OpenCV used for computer vision purposes? How do we use it as a club?
    a. OpenCV includes image detection, constructing 3D images, motion analysis, video editing, and more. UAVs have used OpenCV for image manipulation and feature detection and matching.
3. What are some pre-made alternatives to writing our own aerial imagery code, and how do they work? What are the drawbacks?
    a. ODM (OpenDroneMap) is a program that can take many photos and stitch them together. It is simple to use and implement and can generate 3D models and other data. However, it has too many features that are not needed for UAVs and is not optimized for drone-in-flight.
4. What are the next steps forward as a club?
    a. This year, the club plans to develop a custom image-stitching algorithm that runs in real time on the drone and stitches images into small subcomponents for better efficiency.

Week 3:

1. How can OpenCV and non-NN methods be used for object detection?
    a. OpenCV can be used to isolate properties of an image. Different techniques such as blurring and level curves can be used to reduce noise
2. How are neural networks structured and created?
    a. Neural networks are a large graph of simple functions. Neural networks have nodes that take a set of inputs from previous nodes, apply a weighted activation

function, and send the output to the next set of nodes. Neural networks learn and update by changing the weights of the nodes

3. How do vision models expand on generic neural networks?
    a. Conventional neural networks are an expansion of a simple neural network. They extract features such as shapes and edges. CNNs are stronger for image detection and processing.

Week 4:

1. What is the role of a flight controller in a drone system, and how does it differ from an onboard computer (like a Jetson)?
    a. A flight controller is used mainly to maintain drone stability using sensor data. It processes radio inputs and executes PID control loops such as PIDs. It relies on external components and communities with specific protocols. An onboard computer, such as a Jetson, handles high-level tasks such as autonomy, path planning, and computer vision programs. It sends commands to the flight controller and receives telemetry data for its decision-making.
2. What sensors are typically connected to a flight controller?
    a. Flight controllers are normally connected to an inertial measurement unit, compass, pressure sensor, a GPS module, drone cameras, and RC transmitters and receivers.
3. What are the key communication protocols between flight controllers and peripherals (e.g. I2C, CAN, MAVLink)?
    a. Some of the key communication protocols are:
        i. Serial (UART), a direct basic communication channel between 2 devices
        ii. CAN, a device “group chat.” It is a multi-device protocol used in cars and high-end drones. It is great for long cable runs and unpredictable environments. It also has error checking to prevent miscommunication.
        iii. I2C is used for low-speed communication with devices like magnetometers and IMUs. They share SDA and SCL wires. Multiple devices can talk on the same bus, but only one speaks at a time.
        iv. SPI is similar to Serial but for multiple devices
        v. MAVLink is a message format that runs over UART, UDP, or CAN and is used to send commands and receive data.
        vi. DDS is a protocol used by ROS 2 and drone systems and is great for real-time performance.
4. How does firmware (e.g. Betaflight, PX4) influence the capabilities of the flight controller?
    a. The firmware helps determine what the flight control hardware should actually do.

Week 5:

1. Take notes on nodes, topics, services, actions, QoS, function callbacks
    a. Node:
        i. Does one job (camera, detector, planner)
        ii. Exchange typed messages over topics, services, and actions instead of calling functions.

    iii. Talk in three ways. A topic, service, and action
b. Topics:
    i. Named and typed. Are asynchronous, and publishers don't know who is subscribed or whether anyone is. It never waits for anyone and keeps sending a message.
c. Service:
    i. A client sends a request, and a server sends a response. They are quick, and only one node will serve the request.
d. Action:
    i. In an action, the client sends a goal, and the server accepts or rejects it. The server streams its progress while working on the goal and returns the result of success, if the goal was aborted, or canceled. The client can ask the server to stop the run at any time
e. QoS
    i. QoS is how messages are delivered.
        1. Reliability: resend until it arrives
        2. History Depth: queue the N newest messages; Older ones are dropped.
        3. Durability: you only get messages sent after you subscribe.
f. Callbacks:
    i. A callback is a function you hand to ROS. It is never called by the user. Instead, ROS calls it when a specific event happens, such as receiving a message.